

PCB DESIGN PROJECT REPORT

Design of LM386 Audio Amplifier Circuit

Course / Task: Task 7 - eSim Internship

Software: eSim / KiCad

1. PROJECT OVERVIEW

This project aims to design and layout a Printed Circuit Board (PCB) for an audio amplifier circuit using the LM386 IC. The circuit takes a low-power audio signal as input and amplifies it to drive an output speaker. The schematic capture and PCB layout design were carried out using the eSim EDA software tool.

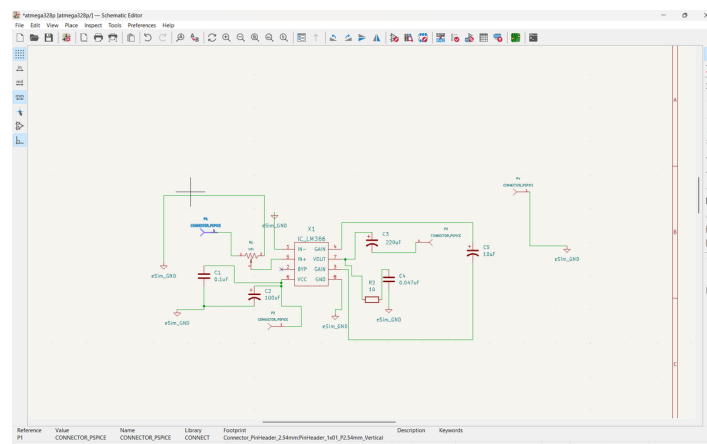
2. CIRCUIT DESCRIPTION

The core component of the circuit is the LM386 low-voltage power amplifier IC. The surrounding passive components are configured to control gain, filter noise, and protect the output:

- **R1 (10k Potentiometer):** Used at the input stage to adjust the volume.
- **C5 (10uF Capacitor):** Connected between pin 1 and pin 8 of the LM386 to set the voltage gain to 200.
- **C1 (0.1uF) and C2 (100uF):** Placed on the power supply line to remove high and low-frequency noise.
- **C3 (220uF Capacitor):** Blocks DC voltage from reaching the speaker while letting the AC audio signal pass through.
- **R2 (10 Ohm) and C4 (0.047uF):** Connected in series at the output pin to form a Zobel network, preventing high-frequency oscillations.
- **P1 to P4:** Standard pin headers for Audio In, VCC, Speaker Out, and Ground connections.

3. SCHEMATIC DIAGRAM

The complete circuit connections were drawn and verified in the Schematic Editor:



4. FOOTPRINT ASSIGNMENT

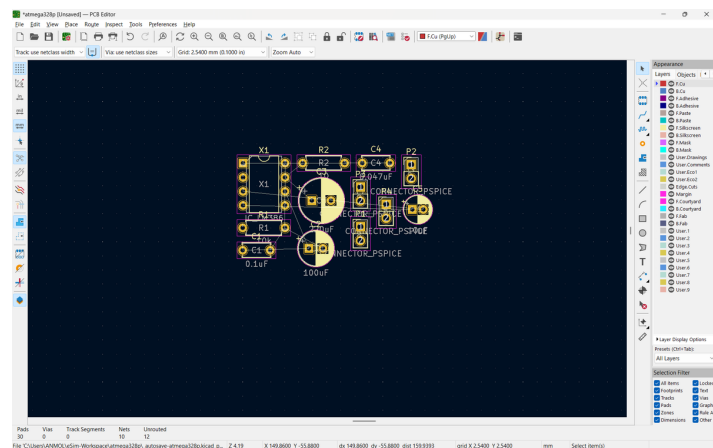
Standard through-hole footprints were assigned to all components for easy manual soldering and assembly:

Component	Value / Description	Assigned Footprint
X1	LM386 DIP-8 IC	Package_DIP:DIP-8_W7.62mm

R1	10k Potentiometer	Potentiometer_THT:Potentiometer_Bourns_3296W_Vertical
R2	10 Ohm Resistor	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
C1, C4	0.1uF, 0.047uF Ceramic Caps	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P2.50mm
C2, C5	100uF, 10uF Electrolytic Caps	Capacitor_THT:CP_Radial_D6.3mm_P2.50mm / D5.0mm
C3	220uF Electrolytic Cap	Capacitor_THT:CP_Radial_D8.0mm_P3.50mm
P1 - P4	1-Pin Connectors	Connector_PinHeader_2.54mm:PinHeader_1x01_P2.54mm_Vertical

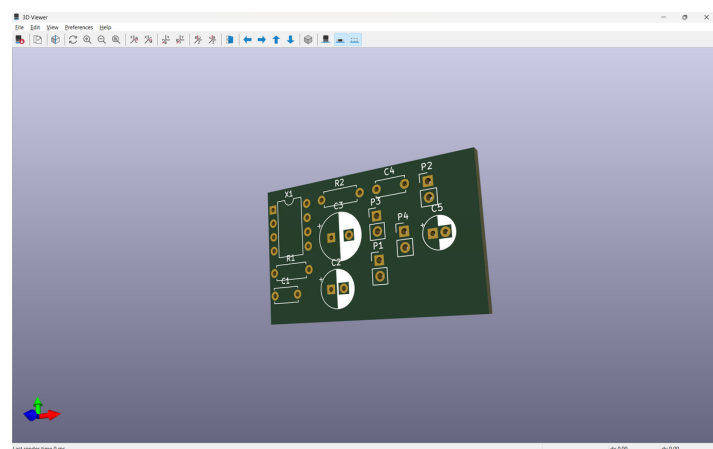
5. PCB LAYOUT AND ROUTING

The netlist generated from the schematic was imported into the PCB layout editor. Components were arranged to keep signal paths short and reduce unwanted interference.



6. 3D VISUALIZATION

The 3D view of the board was checked to verify physical clearance between parts and confirm board dimensions:



7. CONCLUSION

The LM386 audio amplifier circuit was designed and laid out successfully. The schematic and PCB files were generated without errors and are ready for submission.